CH5440 Multivariate Data Analysis

We are drowning in information
and
starved for knowledge

Tom Peters in his book on “Thriving on Chaos”

Harper Perennial, 1991
Edition : Reprint
ISBN-10 : 0060971843
ISBN-13 : 978-0060971847

Resources

Textbooks:

1. Maiti J, Multivariate Statistical Modeling in Engineering and Management, CRC Press, 2023
2. **Backhaus K, B Erichon, S. Gensler, R. Weiber, T. Weiber, Multivariate Analysis – An Application Oriented Introduction, 2nd ed., Springer Gabler, 2023**
3. Johnson R A and D W Wichern, Applied Multivariate Statistical Analysis, Prentice Hall, 2002
4. Montgomery, D C and G C Runger, Applied Statistics and Probability for engineers, Wiley, 2003
5. Joseph F H, W C Black, BJ Babin and R E Anderson, Multivariate Data Analysis, 8th ed., Cengage, 2019
6. Afifi, A., May, S., Donatello, R., & Clark, V.A. (2019). Practical Multivariate Analysis (6th ed.). Chapman and Hall/CRC. https://doi.org/10.1201/9781315203737
7. https://stats.oarc.ucla.edu/other/dae/
8. https://www.routledge.com/Practical-Multivariate-Analysis/Afifi-May-Donatello-Clark/p/book/9781032088471
9. Any reference inadvertently omitted will be added on intimation

Multivariate Data Analysis (MVDA): Introduction

- Analyses of data containing many features (predictors, independent variables) and numerous observations (responses, targets) often running into several thousands

For e.g.,

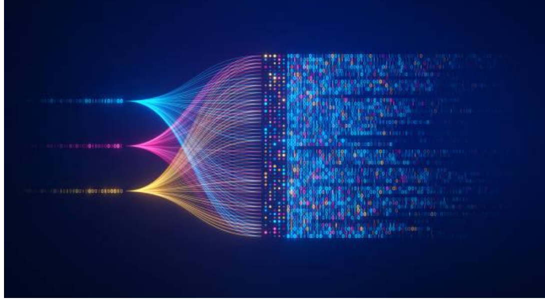
- A medical data base to evaluate liver parameters of young patients
- To evaluate the financial state of a company, the intelligent investor examines 5-10 indexes of the company's performance in that financial year

Multivariate Data Analysis (MVDA): Introduction

- Usually, the responses to at least some of these queries are interrelated/correlated/missing
- Recognizing complicated interrelationships between various features & interpreting these results makes MVDA a highly profitable exercise

Often results and insights are obtained that may not have been attained without multivariate analysis

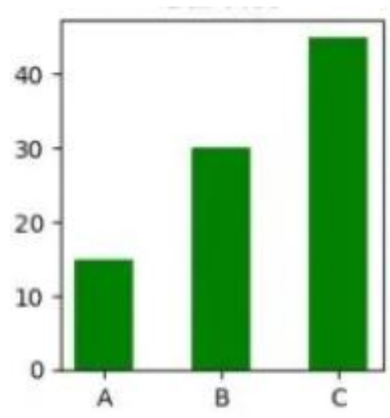
Big Data & Its Management

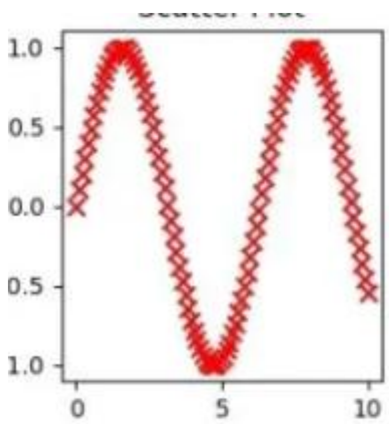


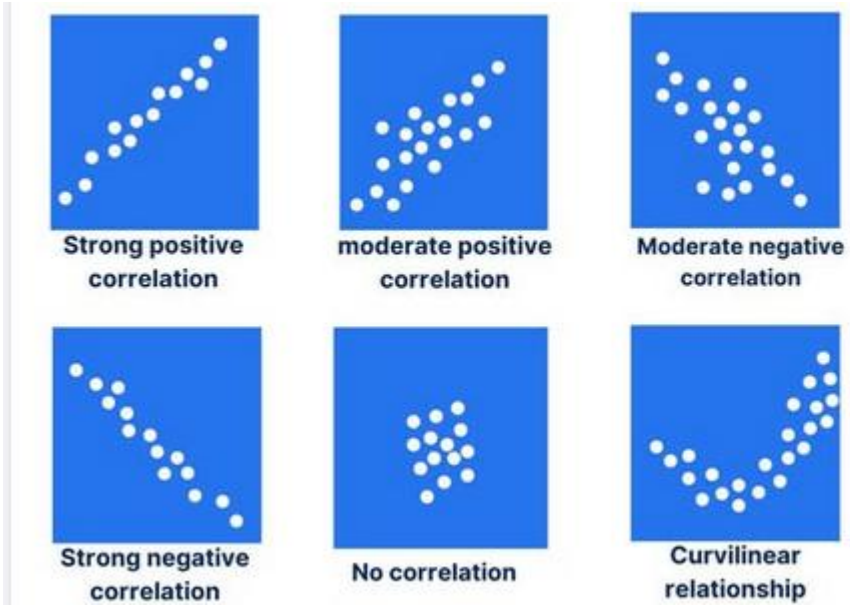
- Explosion in data today
- **Varied Sources:** Social media and online behavioral patterns;
- **IoT** has brought connectivity to almost every type of device;
- Substantial amount of data in genomics, neuroscience and astrophysics;
- Storage devices have capacity to hold Big Data (cloud, servers, data warehouses) and software to manage them
- Companies study and understand their customers through data collected from them/associated with them,
- This data help informed decision-making leading to higher profits

Points to Heed When Collecting Data

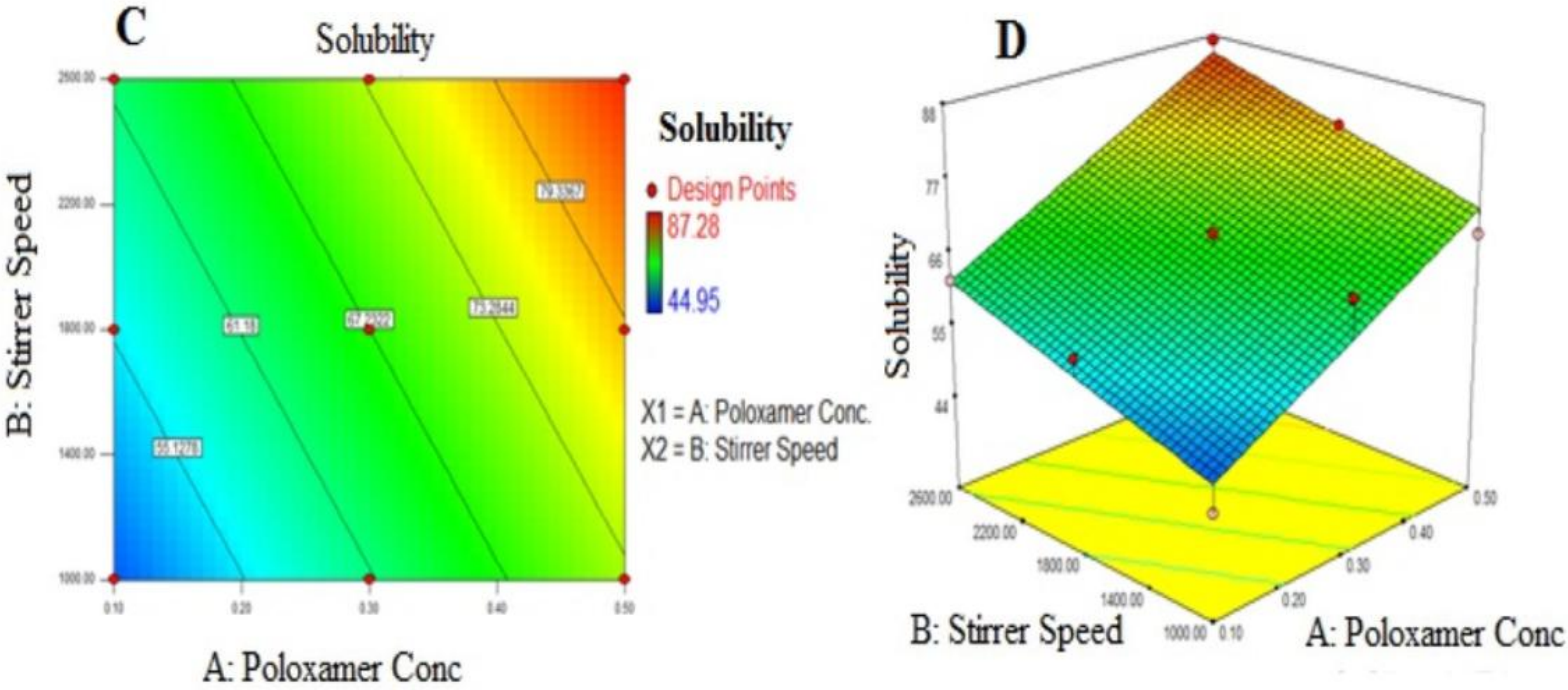
- **Define goals:** Purpose of collecting data
- **Validate :** Data clean, without outliers and missing elements?
- **Analyze:** From the data identify trends, patterns, and correlations
- **Visualize:** Plot the data - box plots, scatter plot matrices, three-dimensional contours
- **Inferring the Data:** How may the information content in the data collected be streamlined to optimized marketing strategies?



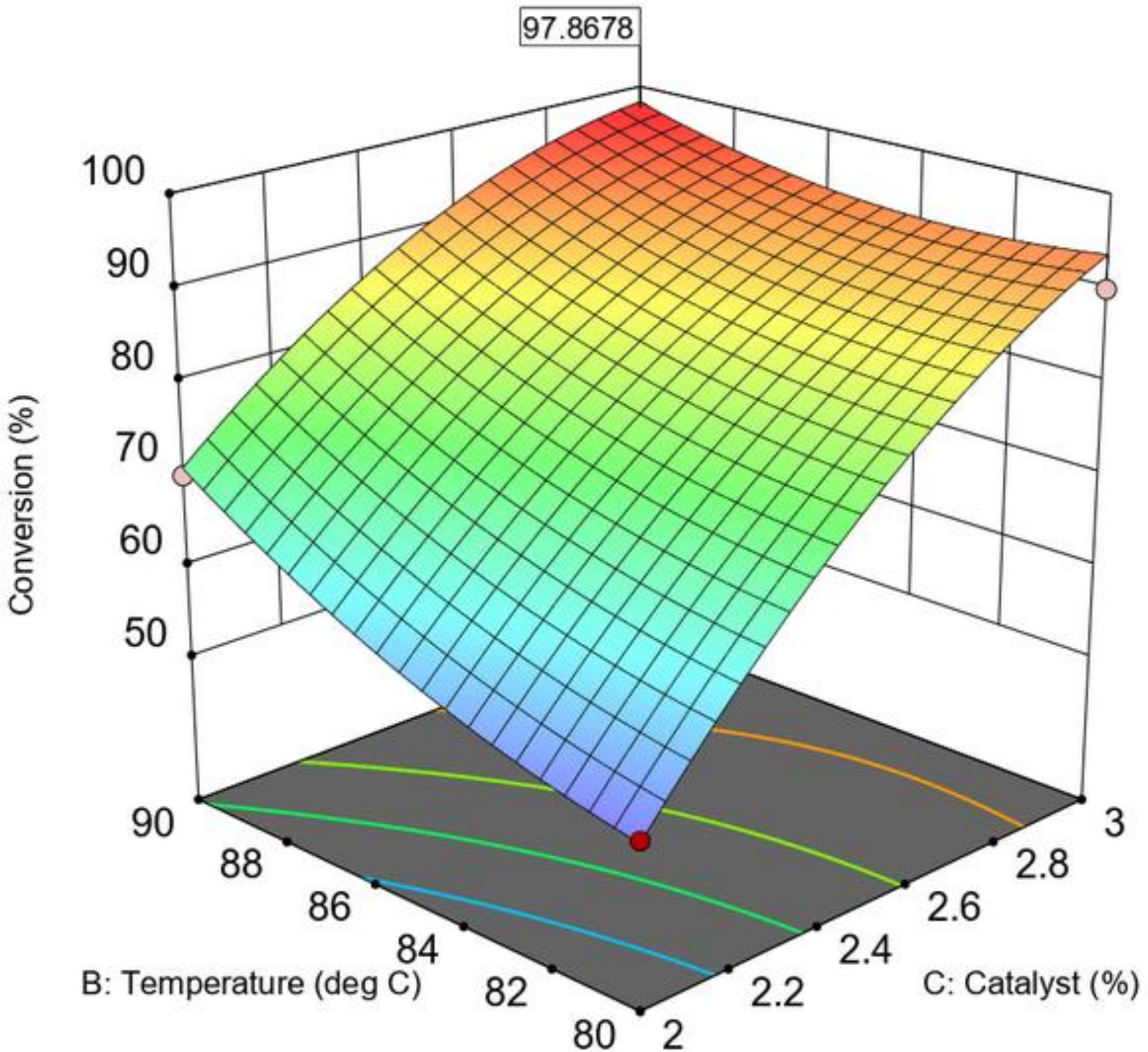




https://medium.com/@boukamchahamdi/key-data-visualization-techniques-for-understanding-trends-and-patterns-1bce981d408f
https://www.facebook.com/AnalyticsVidhya/posts/pearson-correlation-coefficient-r-often-denoted-as-r-measures-the-strength-and-d/967997301998287/



https://www.researchgate.net/figure/Linear-and-3D-response-surface-plot-for-independent-variables-effects-on-the-response_fig3_367231397



n/docs/latest/tutorials/multifactor-rsm/

http://www.scielo.cl/fbpe/img/ejb/v6n2/a09/f1.html last accessed on 23.10.2013

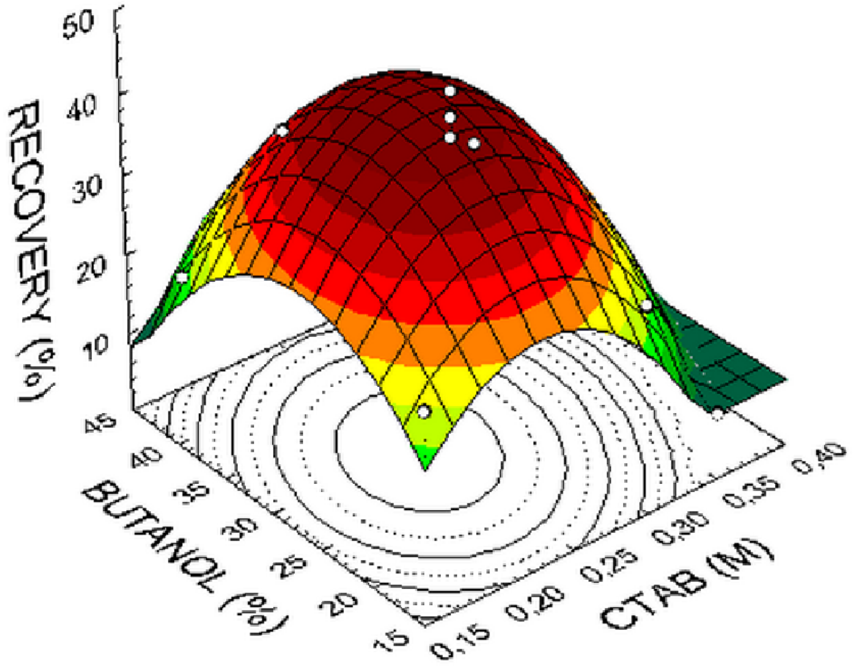
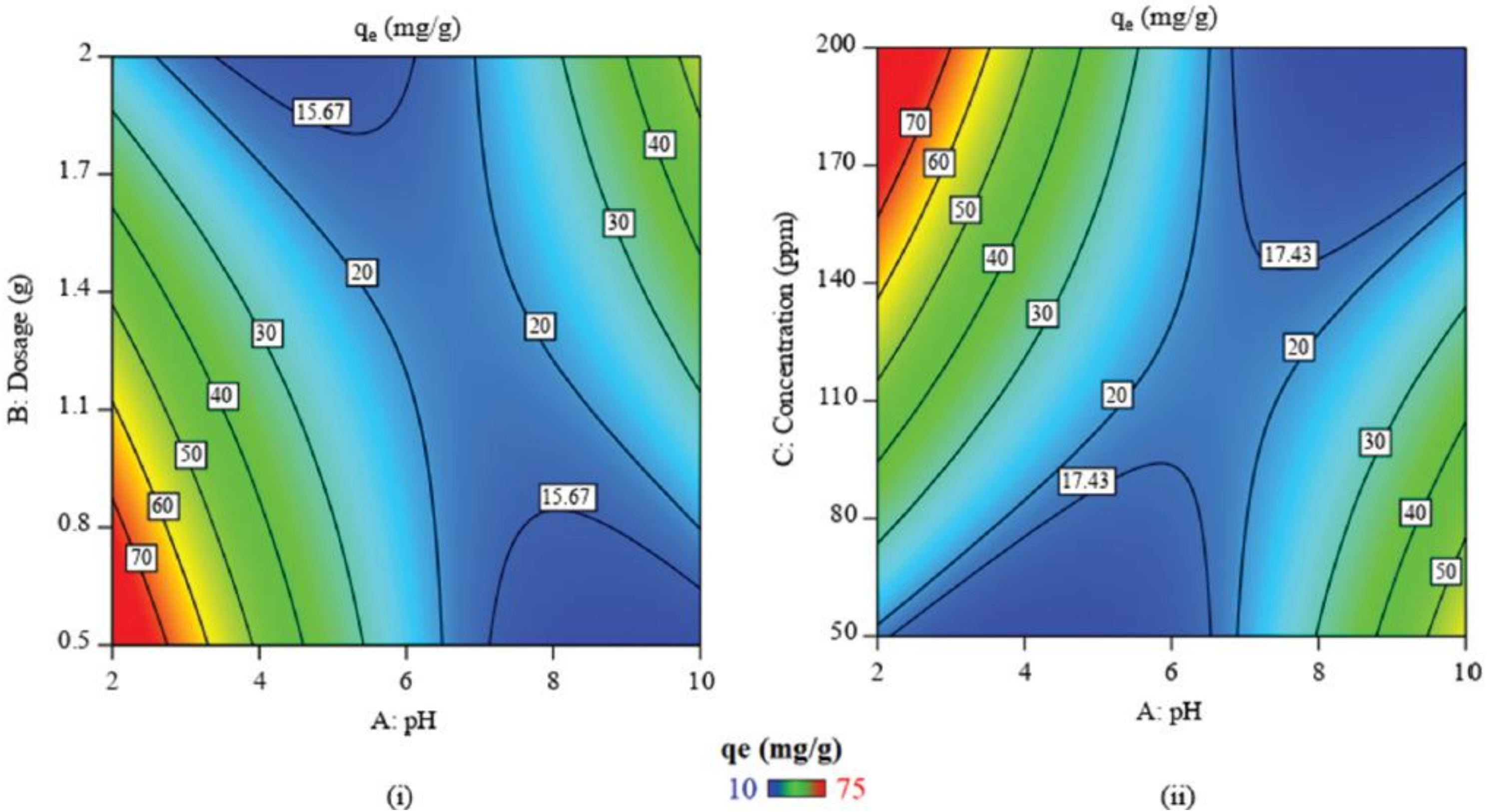
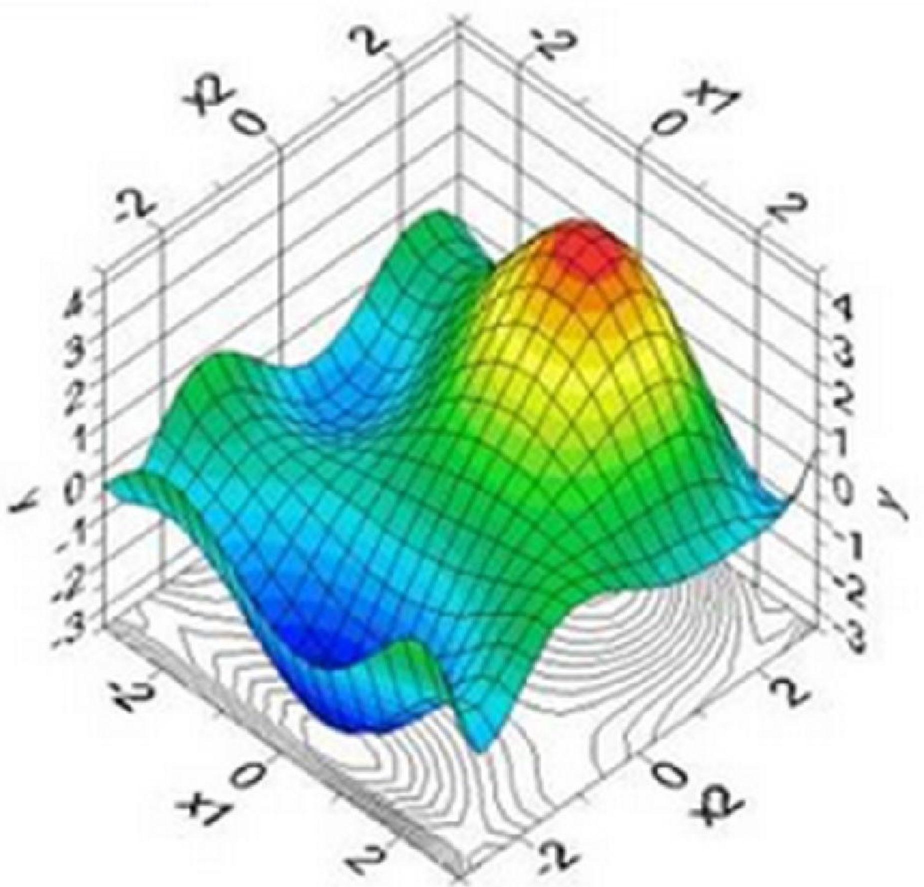


Figure 1. Response surface described by the model, which represents β-xylosidase recovery as a function of CTAB and butanol concentration.

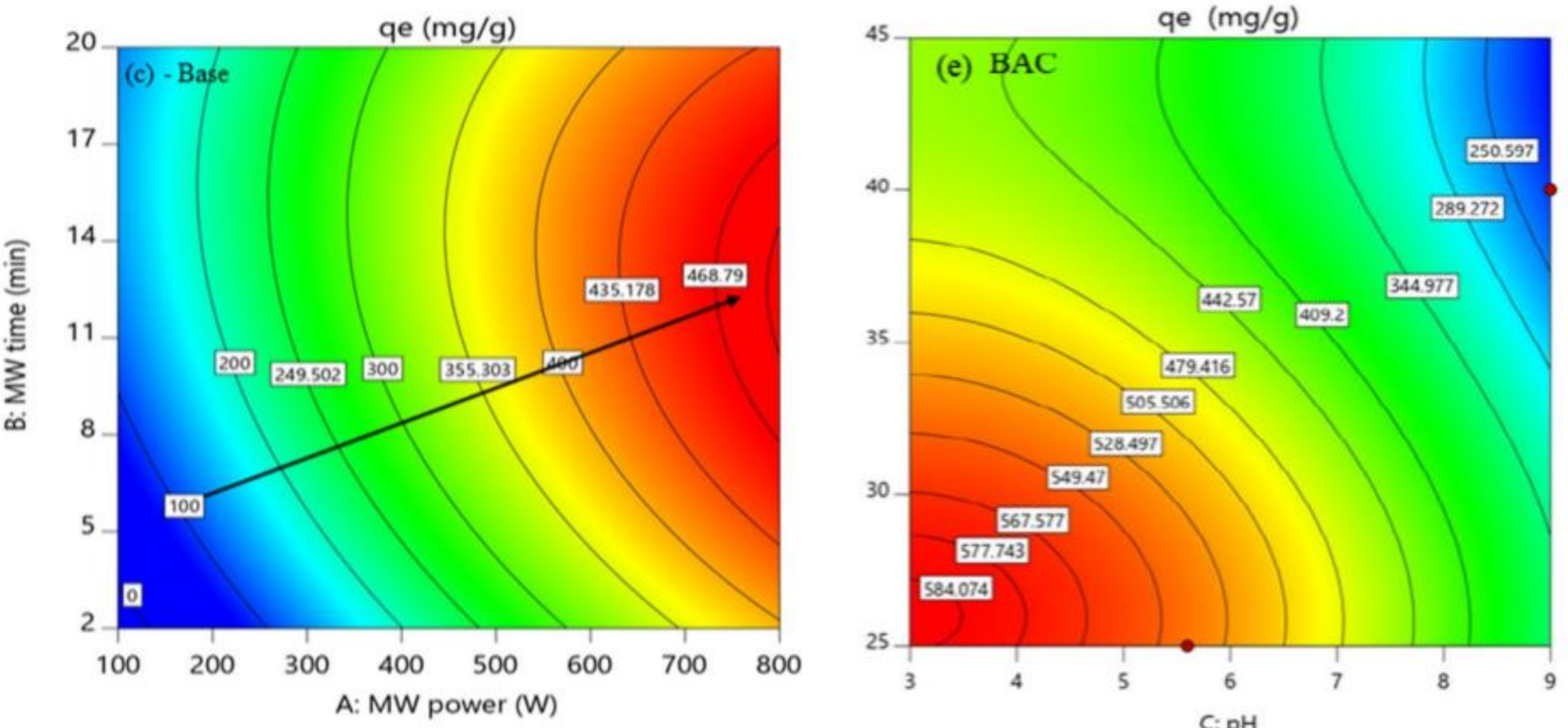
Extraction conditions: pH 8.0, hexanol concentration 10%, electrical conductivity 4 mS/cm, temperature 26�C, CTAB concentration 0.26M and butanol concentration 29.4%.

Presentation of contours, surfaces and 3-D plots





http://www.noesissolutions.com/Noesis/design-optimization/explore/response-surface (last accessed on 23-10-2013)



Environmental Science and Pollution Research (2024) 31:24986–25013

Trends and Patterns

Analyze: From the data identify trends, patterns, and interrelationships

- A trend is the overall direction or long-term movement of a variable over time (e.g., an upward or downward slope).
- Pattern is a specific, repeating design, sequence, or recurring behavior within that data (e.g., daily dips/seasonal spikes or sinusoidal variation)
- A trend dictates the overall direction you are heading, while
- A pattern is the predictable rhythm of events along that journey

Big Data in (Chemical) Engg.

Big Data in (Chemical) Engg.

- Improve productivity, reduce toxic effluents (zero liquid discharge and recycle) and be sustainable
- Facilitate real-time decisions on production scheduling & use of raw materials
- **Preventative maintenance:** Sensors fixed on equipment collect data. Patterns are identified and predictions may be made as to when the equipment might begin to show symptoms of malfunctioning; **Advantage:** Reduced downtime
- **Quality control:** Big data analytics help engineers ensure product quality retention and improvement

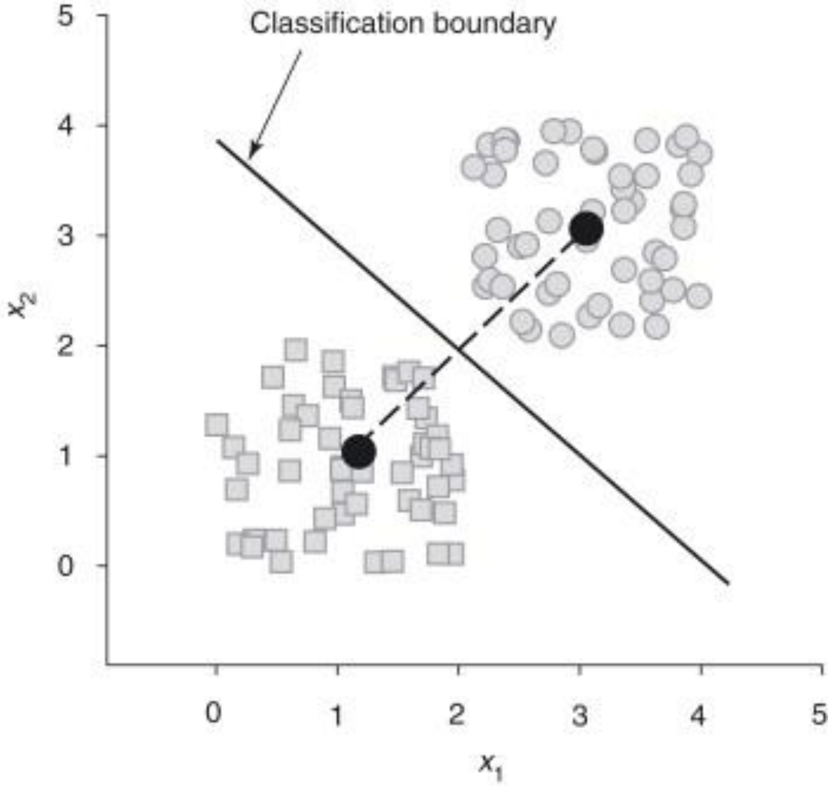
Big Data in (Chemical) Engg.

- **Supply chain management:** Optimize inventory levels
- **Machine learning:** Automate tasks
- **New product development:**
  - New drugs,
  - Alternate pathways for products and processes (new ways of manufacturing), and
  - clean energy solutions

Topics in MVA: Multiple Linear Regression

- An engineer determines variables to maximize the extraction of a medicinal compound from an herb (Speciality Chemical)
- Data on effect of plant leaf size, hot water temperature, hot water amount, time and level of stirring, have been collected.
- A multiple linear regression equation is computed, where % Medicine Extracted is the outcome variable, while the other factors are predictors
- A linear combination of the predictors is used to develop a multivariate linear regression model to predict and optimize the extraction of the herb

Topics : Discriminant Function Analysis (DFA)



Topics : Discriminant Function Analysis (DFA)

- Used when the independent variables are metric
- A method for analyzing differences between two or groups,
- For e.g., how the voting behavior for political parties differs in terms of voters' socio-demographic and psychographic characteristics (attitudes, lifestyles, values, personalities, interests) i/o age and income
- The dependent variable identifies the group membership, that is, the chosen political party, and
- The independent variables describe the group elements, that is, the voters' socio-demographic and psychographic characteristics.

Discriminant Function Analysis (DFA)

- A large sample of initially disease-free men greater than 50 yrs of age from a locality was tracked to see who subsequently suffered from heart attack.
- During the first survey, blood sample was taken from each man, & numerous other tests were conducted - BMI, cholesterol levels, lipids profile, and blood sugar
- Investigator determines a linear function of these to **predict** who would/who would not suffer from a heart attack within the next 10 years. Obviously, these would be a new set of patients.

DFA in Classification

- Once relationships between group membership and characteristics have been analyzed for a given set of subjects
- We may predict the group membership for 'new' subjects

Frequently implemented in

- Credit scoring (i.e., risk classification of bank customers applying for a loan)
- Safe or Unsafe Equipment depending on operating and geometric parameters

Backhaus K, B Erichon, S. Gensler, R. Weiber, T. Weiber, Multivariate Analysis – An Application Oriented Introduction, 2nd ed., Springer Gabler, 2023

Logistic Regression

- Netflix/Prime etc., classifies movies into two different groups according to attraction to a high proportion or only to a low proportion of the viewing audience when the movie is streamed
- The company records data on features such as the movie duration, the genre (action, romance, comedy), and the actors.
- So, variables need not be only numerical but may be categorical as well
- From logistic regression we derive an equation to **estimate the probability** of capturing a high proportion of the target audience

Caution on Models

- If the modeling results are satisfactory (e.g., good model fit, realistic model parameters) the researcher move to the next step
- Make inferences to understand the process being examined
- But researcher must recognize that any conclusions are tied to the proposed model
- Model may not be complete representation/description of the underlying process.

Caution on Models

- If the model incorrectly represents the process, inferences may be flawed (Science is replete with theories that were later disproved)
- Researcher must be aware that there may be many "alternative" models that may also represent the process but provide different conclusions.
- A model is sensitive to the range of features, interaction between them, nonlinearity of the actual model or model parameters drifting, and changing with time

Types of Variables: Manifest Vs. Latent

- Manifest (clear, evident, in plain sight) variables may be directly observed or measured (e.g., weight, height, hair color, age, price, quantities, and income).
- However, latent variables cannot be directly observed
- Examples of hypothetical variables are friendship, rivalry, trust, motivation, intelligence, and potential

Quantifying Latent Variables

- To measure latent variables, suitable operations involving manifest variables are necessary
- **Compositional Approach:** This implies that the value of a latent variable is composed of observations involving various manifest variables
- Intelligence, is often measured by a combination of various observed variables through standardized tests assessing cognitive abilities (reasoning, comprehension, and problem-solving), yielding an IQ score.

Types of Data: Cross-sectional

- Data are the ‘raw material’ to be processed by multivariate data analysis
- Cross-sectional data are collected by observing many different subjects or entities at a single point (snapshot) in time

  (Annual Profits, Type of Service, Years of Operation, Number of Employees of different companies on 31.3.26)

Examples of Cross-sectional Data

- **Comparison** of the financial statements of companies at a fixed date
- **Diabetic levels:** A sample of 1,000 people randomly drawn from a population, with their weight, diet habits and exercise hours measured to calculate the percentage of the sample that is diabetic
- **Price/earnings ratios**: The price-to-earnings ratios of all BSE stocks on a particular date

Cross Sectional Data Spreadsheet

Firm	Annual Salary (Lakhs)	Service Type	Major Gender Hired	HQ Location
A	50	Public Sector	M	North
B	60	Health Care	F	West
C	40	Academics	M	South
D	90	Private Sector	F	East

Hiring of Personnel in 2025

Types of Data: Time Series

- Time Series data (longitudinal data) are collected at regular time intervals (e.g., CO_2 emissions, monthly sales, raw materials intake).
- Time Series Data measure how a variable (d)evolves over time.
- Time series data have a natural temporal ordering

Examples of Time Series Data

- Temperatures and seasonal rainfall measured monthly over a period of many years, will be helpful to predict future weather and climate change and consequently agricultural output
- **Financial data:** Stock prices, interest rates, and industry forecasts published at timely intervals are all examples of financial time series data.

Examples of Time Series Data

- **Sensor data:** Monitoring of heavy machinery to forecast maintenance issues in manufacturing and industrial settings

Real-time data used to identify potential failures before they cause sudden, expensive failures.

Continuously tracking vibration, temperature, pressure etc., enable transition from reactive repairs to proactive maintenance.


PREDICTIVE
MAINTENANCE
FIX BEFORE FAILURE
4:26

Examples of Both Cross-sectional and Time Series Data

Data may be time series as well as cross-sectional when it shows both the characteristics of a group at a specific point in time and also shows the variation in characteristics of that group over a period in time.

Examples:

- Opening prices data of 20 stocks every day for a year is both time series and cross-sectional data.
- Studying the GDP of three developing countries over a period of three years is a combination of time series and cross-sectional data
- 60 Students marks over the multiple evaluations in a semester

Types of Variables: Nominal

- Here, each observation belongs to one of several distinct categories.
- The categories may or may not be necessarily numerical
- For example, “Gender” is a nominal variable.
- An individual’s gender may be male (M) or female (F) category
- Number 07 on MS Dhoni’s vest is a nominal variable (without quantitative value, inherent order, or numerical hierarchy)

Types of Variables: Nominal

- In data analysis, numbers are used as symbols since many computer programs are designed to handle only numerical symbols.
- For example, we may use 0 and 1 to represent males and females, respectively.
- We may also use 1 and 2 to avoid confusing zeros with blanks.
- Any two other numbers can be used provided they are used consistently.

Types of Variables: Nominal

- Different numbers should be assigned to different attributes and identical numbers to identical attributes (one-to-one correspondence).
- For nominal variables, arithmetic operations (such as addition, subtraction, multiplication, or division) are **NOT** permitted
- A mean value of 2.0 for the three numbers 1, 2 and 3 is meaningless.
- However, we can infer frequencies or percentages by counting how often the different attributes occur in a data set (e.g., 25% red; 40% yellow; 35% green).

Types of Variables: Ordinal

- Categories are used for ordinal variables as well, but there also **exists a known order/hierarchy (junior-senior or Fair/Good/ Excellent)** among them.
- Ordinal data contain more information than nominal data, as they reveal order
- Ordinal data are also referred to as rank order data.

Types of Variables: Ordinal

- For e.g., in the Mohs Hardness Scale, minerals and rocks are classified according to ten levels of hardness.
- The hardest mineral is diamond while the softest is talc.
- Any ten numbers may be used to represent the categories, provided they are ordered /sorted in ascending/descending magnitude.
- For instance, the integers 1–10 would be natural to use.
- However, any sequence of increasing numbers may also be used.

Sl. No.	College	Rank
1	X Arts & Science Institute	1
2	Y College of Music	2
3	Z College of Fashion Technology	3
4	S College of Engineering	4

- The best, second best, etc. are obviously evident
- However, the difference between the first and second, the second and third, etc. may not be established
- We see order, but not degree of difference within ordinal data.

Types of Variables: Ordinal

- Often investigators classify people, or ask them to classify themselves, along some continuum
- For example, a physician may classify a patient's disease status as none = 1, mild = 2, moderate = 3, and severe = 4.
- Clearly, increasing numbers indicate increasing severity, but
- It is not clear that the difference between not having an illness and having a mild case is the same as between having a mild case and a moderate one.
- Hence, this is an ordinal variable.

Are months of a Year Nominal?

Treated in many ways depending on what you want to do statistically. As nominal (categorical) variables

- ❖ If you just care about which month (Jan, Feb, Mar, …) with no numerical operations, treat them as nominal.
- ❖ E.G: Comparing average sales across months with dummy variables in regression
- ❖ If your analysis disregards the calendar order and only looks at monthly categories as distinct labels, they may be treated as nominal.

Are months of a Year Ordinal?

- Months of an year considered together is ordinal as there is a definite sequence/order for them (Jan-Dec) – Jan comes before Dec
- If you emphasize order of months (Jan < Feb < … < Dec) but not meaningful numerical distances, treat them as ordinal.
- This may be relevant when you wish to understand "early-, mid-, late-year" patterns

- Best used when evaluating sequential data, rankings, or cumulative events within a single calendar year

Are months of a Year Cyclic?

- Coding Jan = 1, …, Dec = 12 and treating that as plain numeric is usually wrong, because:
- December (12) and January (1) are far apart numerically, but adjacent in time for previous and current cycle.
- For seasonality, months are better modeled as a cyclic variable
- In ML or statistics, months are often transformed using sine and cosine functions to preserve their cyclic nature showing that December and January are close to each other, mathematically

So What is the Deal?

- By default, months are best treated as categorical (nominal).
- They may be treated as ordinal if you just need the ordering.
- Avoid treating 1–12 as a standard numeric/interval scale

Bottom Line: Exercise your Judgement based on Your Problem's Requirement

Types of Variables: Interval

- An interval variable is a variable in which the differences between successive values are always the same
- For e.g., the variable "temperature," in degrees Celsius, is measured on the interval scale since the difference between 12^oC and 13^oC is the same as the difference between 13^oC and 14^oC or the difference between any two successive temperatures
- For these, the value zero like age is just a number

Interval Variables

a) Time Clock   Cyclic

-2 -1 0 1 2 °C

b) T °C or °F

0   -ve values?

c) IQ Scores

d) longitude

e) Calendar Years   2000

Types of Variables: Interval

- In contrast, the ordinal Mohs Hardness Scale does not satisfy this condition since the intervals between successive categories are not necessarily the same.
- The scale must satisfy the basic empirical operation of preserving the **equality of intervals**
- But we can compute average values, standard deviations, or correlations for interval-scaled data.

Types of Variables: Interval

Sl. No.	Month	Temperature (^{o}C)
1	February	30
2	May	45
3	September	36
4	December	25

- Interval scaled data are more informative than nominal or ordinal scaled data, since numbers relate to the quantity.
- Two interval variable type quantities may be directly compared to assess the extent of their difference.
- Example: Temperatures are interval variables.
- Temperature in May is 15^{o}C hotter than in February.
- In December, the temperature is 11^{o}C colder than in September.

Interval Variables: Examples

- ❖ Variables have life below zero in interval variables
- ❖ Overdraft in a bank is interval variable even though currency has a proper zero: Since overdrafts push a balance below zero into the negative territory, it functionally behaves like an interval variable
- ❖ Budget deficits are often treated as interval-scale variables when they can take both positive and negative values.
- ❖ Differences are meaningful: A balance of +200 million ₹ is 300 million ₹ higher than −100 million ₹ .
- ❖ A deficit of −500 million ₹ is 400 million ₹ worse than a deficit of −100 million ₹.
- ❖ The variable can be negative, zero, or positive.

Interval Variables: Examples

More examples are

- ❖ Height with respect to sea-level (20 m above or below sea level).
- ❖ Gauge pressures (can be negative)
- ❖ Heat of reaction (endothermic/exothermic), so are entropy changes (ΔS), Gibbs energy changes (ΔG), internal energy changes (ΔU).
- ❖ Charge (+ve, zero, and –ve) are not interval variables as zero charge implies lack of any charge on the atom – ambiguous
- ❖ Marks in a subject with negative marking is an interval variable

Types of Variables: Ratio

- Ratio variables are interval variables with a reference to “nothing” given by zero .
- For example, height is a ratio variable as 0 height in any dimension is a naturally defined and identified point on the scale.
- We may change the unit of measurement (e.g., cm to feet), but we would still save the zero point and <u>the ratio of any two values of height irrespective of dimensions</u>.

Types of Variables: Ratio

- Temperature is usually not a ratio variable as we may choose the zero point arbitrarily, thus not preserving ratios.
- 0^{o}C is not same temperature as 0^{o}F
- However absolute 0 Kelvin is the baseline for temperature (no -2 K for e.g.,) and is a ratio variable

Types of Variables: Ratio

- Temperature is always not a ratio variable as we may choose the zero point arbitrarily, thus not preserving ratios.
- 0°C is not same temperature as 0°F.

Variable Name	Ratio	Value (Interpretation)
Height (m)	10 m /5 m	2 (twice as tall)
Height (cm)	1000 cm / 500 cm	2 (twice as tall)
Temperature (°K)	10°K/5°K	2 (twice as hot)
Temperature (°C)	10°C/5°C	2 but NOT twice as hot

Types of Variables: Ratio

- Temperature is always not a ratio variable as we may choose the zero point arbitrarily, thus not preserving ratios
- $10^oC/5^oC$ in Kelvin temperature (a ratio variable) units becomes 283.15/278.15 = 1.018 which is not the same as 2
- 10 bar(g)/5 bar(g) =2 is not same as 11 bar(abs)/6 bar (abs)
- Pressure In bar(g) (or temperature in ^oC or ^oF) do not share the SAME zero point while height (in any units) and weight (in any units) have their unique zero points

Absolute and Relative Zeros

- Height and weight have absolute, undeniable zero points representing the total absence of length or mass.
- Conversely, bar(g) & many temperature scales utilize relative zeroes.
- Their zero points are arbitrary references set for human convenience, which is why temperature can go "below zero."
- In 0 bar(g) or 0 ^{o}C or 0 ^{o}F we do not have complete absence of pressure or temperature

Types of Variables: Ratio

- ❖ Ratio variables need the reference to zero below which there is no other number/value
- ❖ A defining feature of a ratio scale is that it has a true (absolute) zero, so ratios are meaningful as they are measured relative to that zero point
- ❖ Variables which may take negative values are interval variables but not ratio variables

Types of Variables: Ratio

- Ratio-scaled data permits all arithmetic operations as well as the calculation of all statistical measures
- Size of a swelling is a ratio variable as zero size indicates complete absence of swelling
- However when a Doctor classifies swelling as 0-none, 1-moderate, 2-severe then it becomes an ordinal variable

Types of Variables: Ratio

- Ratio scaled data convey the greatest amount of information, as ‘0’ has meaning.
- Value of zero indicates complete absence of the entity being measured.
- With a true zero in your scale, you can calculate ratios of values.
- E.g., you can say that 4 children is twice as many as 2 children in a household.
- Similarly, 8 years is double 4 years of experience.
- Business variables measured in $, such as revenues and profits, as well as those measured in percentages, such as market share, are ratio scaled variables.
- A value of zero means no revenue, no profits, or no market share.

Types of Variables: Binary & Dummy

- Variables that comprise only **2** categories are termed dichotomous
- Special case of nominal variables. For e.g.: YES/NO decisions (e.g., purchase/non-purchase), positive/negative test results (e.g., Corona-positive/Corona-negative), question of survival (e.g., patient survived/patient died), tossing a coin (head/tail), pass/fail
- Dichotomous variables are mutually exclusive, occurrence of one precludes the occurrence of the other

Types of Variables: Binary & Dummy

- Occurrence of one instance of the binary dichotomous variable precludes the occurrence of the other – If YES, NO is ruled out.
- If dichotomous variables are coded as 0 and 1, they are called binary variables or dummy variables.
- Binary variables may be treated like metric variables.

Types of Variables: Binary & Dummy

- The mean value of a binary variable indicates the proportion the attribute value coded with 1 occurs in a data set

$$\frac{\sum_{i=0}^{n} \text{number of ones}}{n}$$

- E.g., ‘purchase’ is coded with 1 and ‘no-purchase’ with 0, the mean value of 0.75 means that the product was purchased by 75% i.e., by 30 out of the 40 respondents.

Types of Variables: Dummy

- Nominal variables with more than two categories may not be treated like metric variables.
- But it is possible to replace a nominal variable say “green” among three colors “red”, “green” and “blue” with several dummy variables that may then be interpreted like metric variables (say 1 0)

$$\begin{array}{ll} \text{Red} & \\ \text{Green} & \\ \text{Blue} & \end{array} \begin{Bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{Bmatrix} \text{(Ref)} \qquad \begin{array}{ccc} \multicolumn{3}{c}{\text{One hot encoded}} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}$$

Types of Variables: Dummy

- Supplier wants to know if customer choice is linked to color of packing
- Three packing colors red, yellow and green are considered
- The nominal variable ‘color’ may be represented by three dummy variables (q_1, q_2, and q_3), each of which may have the value 1 (color present) or 0 (otherwise).
- For color red, for example, the coding of the dummy variable q_1 is as follows:

$$q_1 = \begin{cases} 1 & \text{if color} = \text{red} \\ 0 & \text{otherwise} \end{cases}$$

Types of Variables: Dummy

- Similarly, a dummy variable q_2 maybe defined for yellow and a dummy variable q_3 for green.
- So, we need not do 3-levels [1 0 0], [0 1 0] and [0 0 1] but work with only 2 levels
- The third variable [0 0] serves as the reference category and is definitely one of the possibilities in the data set

Types of Variables: Dummy

- The choice of the reference category is arbitrary.
- In general, a nominal variable with n categories may be replaced with (n−1) binary columns
- Dummy variables representation is different from one – hot encoding representation

Difference Between One-Hot Encoding and Dummy Variables

Dummy variables and one-hot encoded variables are both techniques used to convert categorical text data into numbers, but they have a key mathematical difference: **One-hot encoding creates a new column for every category, while dummy encoding creates $N-1$ columns** (where N is the total number of categories). Data Science Stack Exchange +2

d1	d2	d3
1	0	0
0	1	0
0	0	1

A simple breakdown illustrates how they handle a color category with 3 options: Red, Blue, and Green.

The Core Differences

Feature	One-Hot Encoding	Dummy Encoding
Number of columns created	All N categories are represented by N binary columns. binary column	$N-1$ columns are created. One ~~category~~ is omitted ~~and~~ serves as the baseline. ↳ The third [0 0] category serves as baseline
Example for Colors (Red, Blue, Green)	Requires 3 columns: *Color_Red*, *Color_Blue*, *Color_Green*.	Requires 2 columns: *Color_Blue*, *Color_Green*. (Red is represented by `[0, 0]`).

Relative merits and Demerits

Merits and Demerits

Feature	One-Hot Encoding	Dummy Variable Encoding
Number of variables	k Binary variable contg. column	k−1 k − 1 binary variable containing columns
Information retained	Complete	Complete (reference category implied)
Multicollinearity	Causes perfect multicollinearity if intercept is included	Avoids dummy-variable trap
Memory requirement	Higher	Lower
Computational cost	Higher	Lower
Interpretation	Symmetric treatment of all categories	Interpretation relative to baseline category
Suitable for tree models	Excellent	Excellent
Suitable for linear/logistic regression	Needs care (drop one column or remove intercept)	Preferred
Coefficient interpretation	Difficult due to redundancy	Easier

$X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\boxed{\hat\beta_0}$

$\hat{y}_{CH} = \hat\beta_0 + \hat\beta_1 x_1 + \hat\beta_2 x_2$

$\hat{y}_{ML} = \hat\beta_1 x_1 + \hat\beta_2 x_2$

With Intercept

- One-Hot Encoding can create multi-colinearity with intercept included

```
Data = [
1 0 0 3
0 1 0 5
0 0 1 11
1 0 0 9
0 1 0 12
];
Y=[5 8 15 13 18];
X= [ones(numel(Y),1) Data(:,1) Data(:,2) Data(:,3) Data(:,4)];
beta_hat = (inv(X'*X))*(X'*Y);

X =
1 1 0 0 3
1 0 1 0 5
1 0 0 1 11
1 1 0 0 9
1 0 1 0 12
```

Can we say columns are linearly Independent?

$(X'X)^{-1}$

```
(X'*X) % Linearly independent columns?

% 5 2 2 1 40
% 2 2 0 0 12
% 2 0 2 0 17
% 1 0 0 1 11
% 40 12 17 11 380
% First column is the sum of the remaining columns

% 1.0e+15 * BLOWING UP
%
% 9.3801 -9.3801 -9.3801 -9.3801 0.0000
% -9.3801 9.3801 9.3801 9.3801 -0.0000
% -9.3801 9.3801 9.3801 9.3801 -0.0000
% -9.3801 9.3801 9.3801 9.3801 -0.0000
% 0.0000 -0.0000 -0.0000 -0.0000 0.0000

size(X'*X) = 5 5
rank(X'*X) = 4

beta_hat =
 -0.0263
  1.9557
-30.5737
 64.8969
  1.3882
```

(5×5)

rank $\neq 5$

$(X'X)^{-1} = 10^{15} \times$

$4 < 5$

With Intercept - PY

- One-Hot Encoding can create multi-colinearity with intercept

```
import numpy as np

 Data = np.array([ [1, 0, 0, 3], [0, 1, 0, 5], [0, 0, 1, 11], [1, 0, 0, 9], [0, 1, 0, 12] ])
Y = np.array([5, 8, 15, 13, 18]).reshape(-1, 1)
X = np.hstack((np.ones((len(Y), 1)), Data))
XtX = X.T @ X
beta_hat = np.linalg.pinv(XtX) @ (X.T @ Y)
```

X =

1	1	0	0	3
1	0	1	0	5
1	0	0	1	11
1	1	0	0	9
1	0	1	0	12

XtX # Linearly independent columns?

5	2	2	1	40
2	2	0	0	12
2	0	2	0	17
1	0	0	1	11
40	12	17	11	380

First column is the sum of the remaining columns
1.0e+15 * BLOWING UP

9.3801	-9.3801	-9.3801	-9.3801	0.0000
-9.3801	9.3801	9.3801	9.3801	-0.0000
-9.3801	9.3801	9.3801	9.3801	-0.0000
-9.3801	9.3801	9.3801	9.3801	-0.0000
0.0000	-0.0000	-0.0000	-0.0000	0.0000

XtX.shape
(5,5)
np.linalg.matrix_rank(XtX)
4 < 5

$\mathrm{Var}(\hat{\beta}) = (X'X)^{-1}\hat{\sigma}^2$

beta_hat =

$$\left\| \begin{matrix} -0.0263 \\ 1.9557 \\ -30.5737 \\ 64.8969 \\ 1.3882 \end{matrix} \right\|$$

Precision of parameters
Very Poor

Without Intercept

- No multicolinearity without intercept

```
clc; clear all
Data = [
1 0 0 3
0 1 0 5
0 0 1 11
1 0 0 9
0 1 0 12
];
Y=[5 8 15 13 18]';
X=[Data(:,1) Data(:,2) Data(:,3) Data(:,4)];

% No ones column (no intercept)

% X =
%
% 1 0 0 3
% 0 1 0 5
% 0 0 1 11
% 1 0 0 9
% 0 1 0 12
```

no ones

```
beta_hat = (inv(X'*X))*(X'*Y);
% beta_hat =
% 0.6706
% 1.2000
% -0.2706
% 1.3882

(X'*X) % Linearly independent columns?
% 2 0 0 12
% 0 2 0 17
% 0 0 1 11
% 12 17 11 380
```

4 x 4

```
% First column is NOT the sum of the remaining columns
inv(X'*X)
% 1.3471 1.2000 1.5529 -0.1412
% 1.2000 2.2000 2.2000 -0.2000
% 1.5529 2.2000 3.8471 -0.2588
% -0.1412 -0.2000 -0.2588 0.0235
size(X'*X) % (4,4)
rank(X'*X) % 4 Full
```

Full rank

Rank = 4

X matrix = 4 x 4

Without Intercept - PY

- No multicolinearity without intercept

```
import numpy as np

 Data = np.array([ [1, 0, 0, 3], [0, 1, 0, 5],
[0, 0, 1, 11], [1, 0, 0, 9], [0, 1, 0, 12] ])
Y = np.array([5, 8, 15, 13, 18]).reshape(-1,
1)
X = Data
XtX = X.T @ X
beta_hat = np.linalg.pinv(XtX) @ (X.T @
Y)

% No ones column (no intercept)

% X =
%
% 1 0 0 3
% 0 1 0 5
% 0 0 1 11
% 1 0 0 9
% 0 1 0 12
```

```
beta_hat
0.6706
1.2000
-0.2706
1.3882

XtX % Linearly independent columns?
2 0 0 12
0 2 0 17
0 0 1 11
12 17 11 380

# First column is NOT the sum of the remaining columns
np.linalg.pinv(XtX)
1.3471 1.2000 1.5529 -0.1412
1.2000 2.2000 2.2000 -0.2000
1.5529 2.2000 3.8471 -0.2588
-0.1412 -0.2000 -0.2588 0.0235
XtX.shape
( 4,4)
np.linalg.matrix_rank(XtX)
4
```

What is Dummy Variable Trap?

With one-hot encoding and an intercept:

$$Red+Green+Blue=1$$

One hot encoded value for Green = 1 − Σ one hot encoded values of other colors.

- ❖ The columns become linearly dependent.
- ❖ For example, $Blue=1-Red-Green$
- ❖ This causes perfect multicollinearity, making the design matrix singular in ordinary linear regression.
- ❖ Dummy variable encoding avoids this problem by dropping one category.

What is Dummy Variable Trap?

When a categorical variable with k categories is represented using **one-hot encoding**, the k indicator columns always satisfy

$$x_1 + x_2 + \cdots + x_k = 1$$

for every observation, because exactly one category is present.

If you also include an intercept column, the intercept column is

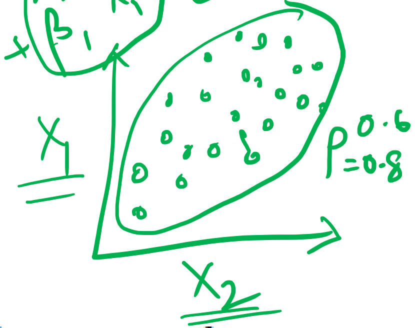
$$1 = x_1 + x_2 + \cdots + x_k$$

which means the intercept is an exact linear combination of the one-hot columns. This creates **perfect multicollinearity** and the regression design matrix becomes singular.

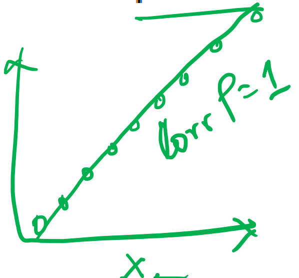
regression

$$\hat{y} = \hat{\beta}_0 + \boxed{\hat{\beta}_1 x_1} + \boxed{\hat{\beta}_2 x_2}$$

0.5 ~0



$(X'X)^{-1}$ blows



Types of Variables: Indicator

- The usual method of representing the different levels of a qualitative variable is by using indicator variables.
- For example, to introduce the effect of two different machines into a regression model, we could define an indicator variable as follows:

$$x = 0 \quad \text{if the observation is from machine 1}$$
$$x = 1 \quad \text{if the observation is from machine 2}$$

Collinearity

$$Var(\hat{\beta}) = (X'X)^{-1}\,\boxed{\hat{\sigma}^2}$$

$$\hat{\sigma}^2 = \frac{SS_{Residuals}}{n-p}$$

error

- In general, a qualitative variable with t levels is represented by t−1 indicator variables, which are assigned the values either 0 or 1
- Indicator and dummy variable are one and the same or are they?

Difference Between Indicator and Dummy Variable

Example 1: Indicator variable

Suppose a reactor is operating.

$$I = \begin{cases} 1 & \text{if pressure exceeds safety limit} \\ 0 & \text{otherwise} \end{cases}$$

This is an indicator variable because it indicates whether a condition is met.

Example 2: Dummy variables

Suppose Catalyst Type has three categories: A, B, and C.

Catalyst	D_1	D_2
A	1	0
B	0	1
C	0	0

Here D_1 and D_2 are dummy variables used in a regression model.

Mathematical view

A variable of the form

$$I_A = \begin{cases} 1 & \text{if event } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

is called an **indicator variable** in probability theory.

Difference Between Indicator and Dummy Variable

If yes/no then indicator variable and if blue/green then dummy variable

In statistical modeling and machine learning, you are describing the process of converting categorical data into formats algorithms can process:

- **Yes/No (Indicator Variables):** Also known as a **binary variable**, this takes on only two possible values, typically 1 for "Yes" (presence of a trait) and 0 for "No" (absence of a trait).
- **Blue/Green (Dummy Variables):** When a variable has more than two categories (e.g., Red, Blue, Green), you create a separate dummy variable for all but one of the categories. The unrepresented category becomes the baseline or "reference" group.

Types of Variables: Indicator & Dummy

❖ Thus, if there were three machines, the different levels would be accounted for by two indicator variables defined as follows:

x_1	x_2	
0	0	if the observation is from machine 1
1	0	if the observation is from machine 2
0	1	if the observation is from machine 3

Indicator variables are also referred to as dummy variables.

Minor Semantic Distinction

While the terms are synonymous, statisticians sometimes make a slight distinction in how they refer to **groups**:

- **Indicator Variable**: Typically used to describe an isolated binary condition (e.g., Yes/No).
- **Dummy Variable**: Often used when a single categorical variable has 3 or more levels and must be broken down into a set of indicators. For example, a "Day of the Week" variable would be represented by a set of dummy variables (e.g., Monday = 1 , others = 0). Stack Exchange +2

Types of Variables: Discrete and Continuous

- Many authors use the term categorical to refer to nominal and ordinal variables
- Variables may be classified as discrete or continuous.
- A variable is called continuous if it can take on any value in a specified range.
- Thus, the height of an individual may be 70 or 70.4539 inches.
- Any numerical value within a certain range is a possible height.

Types of Variables: Discrete and Continuous

- A variable that is not continuous is discrete.
- A discrete variable takes on only certain specified values but nothing between the chosen ones are considered
- For e.g., counts are discrete variables since only 0 or positive integers are allowed.
- **In fact, all nominal and ordinal variables are discrete**.
- Interval and ratio variables may be continuous or discrete.

Types of Variables: Discrete and Continuous

- Nominal & ordinal are non-metric/categorical scales
- Interval and the ratio are metric or cardinal scales
- Scale decides information content of the data & permissible math and statistical operations
- Nominal scale is the lowest & ratio scale the highest in terms of information content
- Permissible arithmetic operations or statistics that are permissible on a lower scale are also permissible on a higher scale, but not vice versa

Types of Variables: Discrete and Continuous

- Superior the measurement level, the more is the information content of the data
- More arithmetic and statistical operations may be performed on the data.
- Possible to transform data from a higher to lower scale, but not vice versa.
- May be useful for increasing the clarity of the data or for simplifying the analysis
- For example, in many cases income classes or price classes are formed.
- This may involve transforming the originally ratio-scaled data to an interval, ordinal or nominal scale.
- Of course, transformation to a lower scale always involves a loss of information

Types of Operations Permitted

Type of Variable	Constitutent	Permitted Math/Stat Operations
Categorical	Nominal	Frequency, Mode
	Ordinal	Median, Quantiles
Metric	Interval	Difference, Average, Standard Deviation, Correlation
	Ratio	Total, ratio, multiplication, geometric and harmonic averages, and coff. of variation

1. Backhaus K, B Erichon, S. Gensler, R. Weiber, T. Weiber, Multivariate Analysis – An Application Oriented Introduction, 2nd ed., Springer Gabler, 2023